

ALEX CHEVRIER — Open to Low-Latency C++/FPGA Role
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SKILLS

C++23, Java 25, SQL | lock-free SPSC/MPSC, seqlock, mmap, DPDK, memory ordering, pthread_setaffinity_np | Vitis HLS, RTL co-simulation (xsim) | GBench/GTest, perf stat, CMake/Meson, Conan | Linux/GCC, Kafka, Docker, AWS

PROJECTS

FPGA Feed Handler (Vitis HLS/C++) | github.com/alchevrier/fpga-feed-handler

ITCH 5.0 parser → AOS BRAM order book → snapshot registers; 4-stage DATAFLOW pipeline. Dual-feed MOLDUDP64 arbitration in hardware; compliance window zero by architecture.

CoSim verified on Artix-7 (250 MHz): 13 cycles / 52 ns, II=16. Arbitration cost +4 cycles vs single-feed baseline (both CoSim verified). 11 ADRs.

Low-Latency Feed Handler (C++23) | github.com/alchevrier/low-latency-feed-handler

ITCH 5.0 zero-copy parser → MPSC → SOA order book; DPDK pcap PMD. Zero heap end-to-end.

163M real NASDAQ messages (8,669 symbols): P99.9 274 ns parse + book insert; 97.5% branch accuracy.

Low-Latency Order Book (C++23) | github.com/alchevrier/low-latency-order-book

N-SPSC MPSC, SOA order book, seqlock, pinned threads. P50 2.5 ns isolated; P99.9 6.3 µs under cross-core write pressure. 10 ADRs.

EXPERIENCE

Morgan Stanley, Hong Kong — Software Engineer (Contract) Nov 2025–Present

- Designed greenfield messaging service; reverse-engineered Python batch system into streaming Kafka pipeline.
- Maintain production Kafka infrastructure across isolated data centres.

We+, Hong Kong — Software Engineer Sept 2023–Nov 2025

- Migrated highest number of microservices to Java 21 across French investment bank FX platform.

LandyTech, London — Software Engineer Oct 2019–Aug 2023

- Monolith-to-microservices migration; automated risk data pipeline.

Redbite Solutions, Cambridge — Software Engineer Nov 2016–Oct 2019

- Built web application from scratch; implemented OAuth2 authentication layer.

EDUCATION

ENSEIRB-MATMECA, Bordeaux — M.Sc. Computer Science 2013–2016

Admitted by competitive national examination (Grande École entrance).